

東京 2025 LINUX PLUMBERS CONFERENCE

TOKYO, JAPAN / DECEMBER 11-13, 2025

Android Boot: Next Steps

Dmitrii Merkurev <dimorinny@google.com>



東京 2025
LINUX
PLUMBERS CONFERENCE
TOKYO, JAPAN / DEC. 11-13, 2025

GBL recap

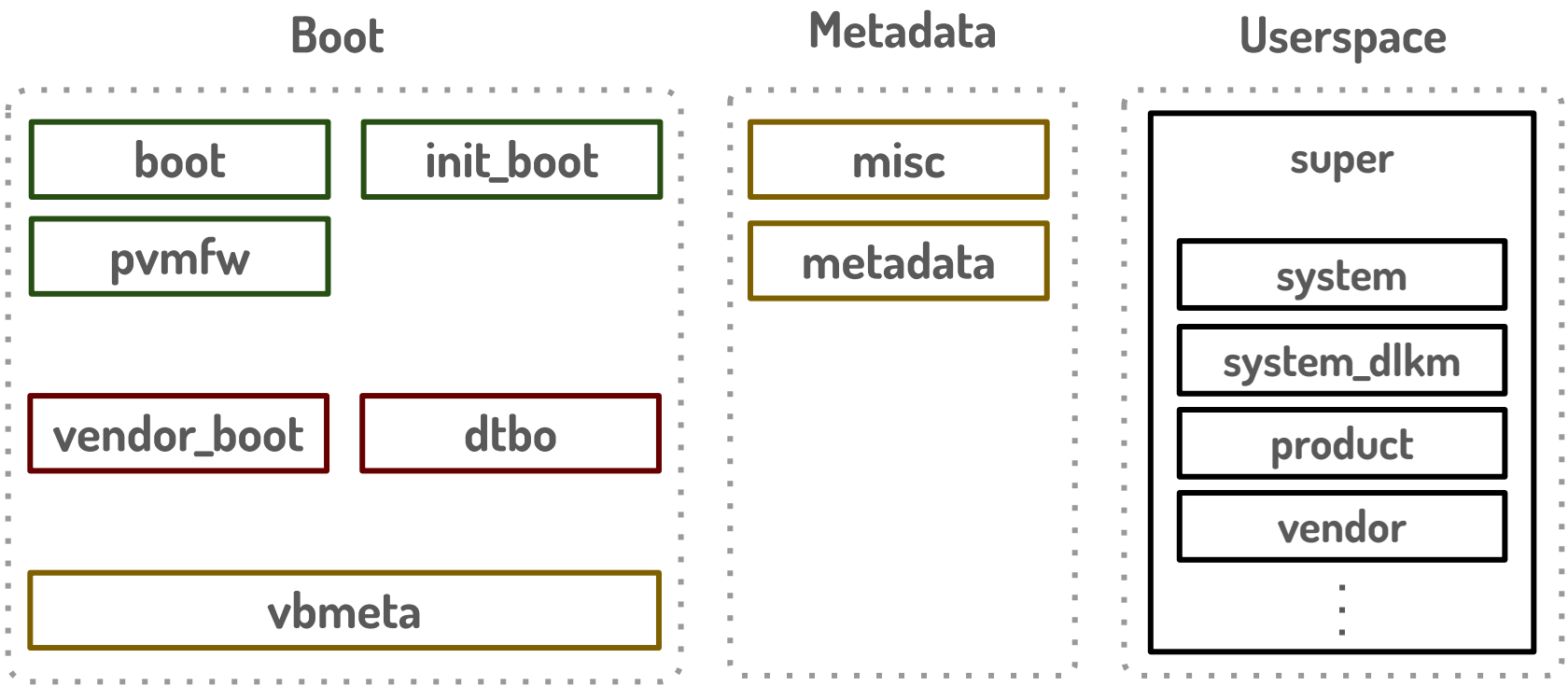
Generic bootloader (GBL) is Android boot flow UEFI application provided by Google.

Objective: Provide common Android boot implementation so we can:

1. Reduce the vendor's integration burden
2. **Ensure we can modernize boot flow to support new features, achieve better performance, security, alignment with the rest of the ecosystem.**

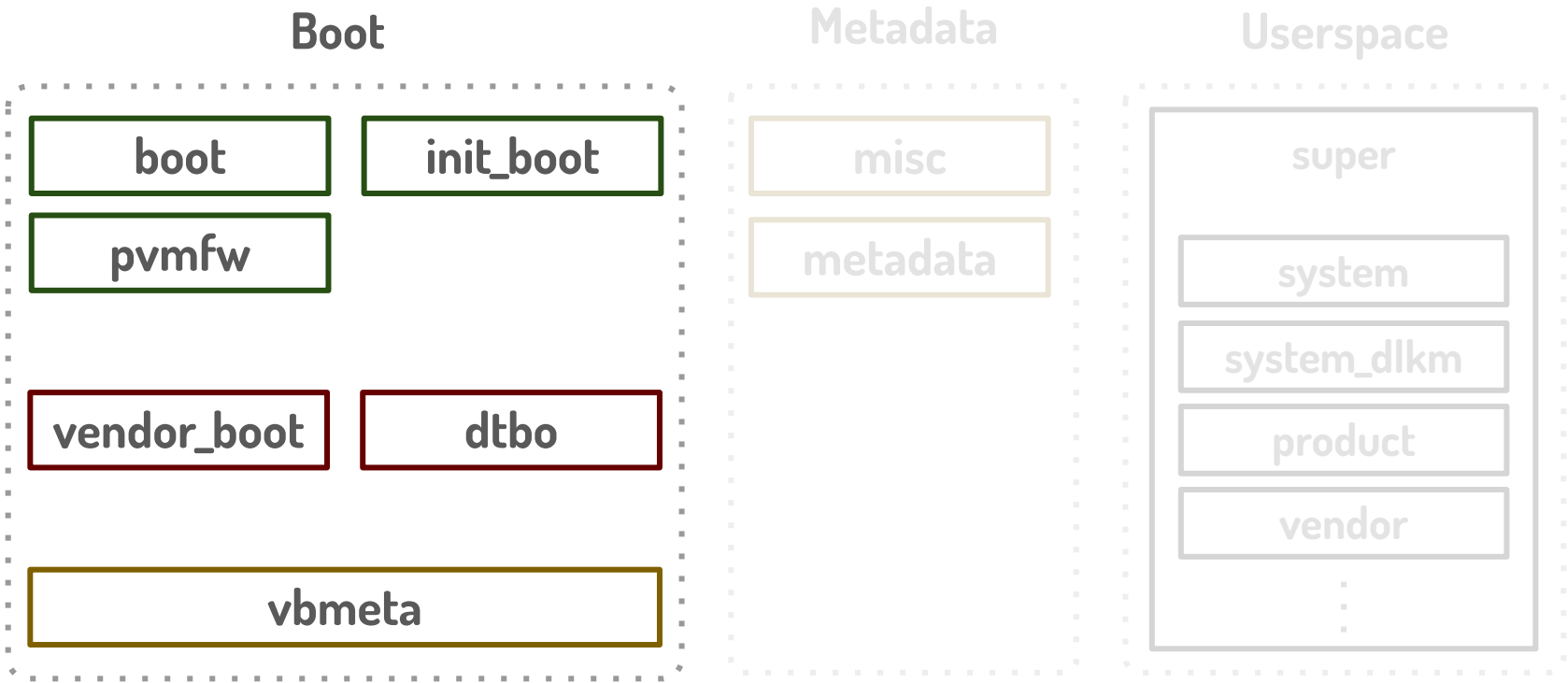
Without GBL p.2 is expensive.

Android partitions



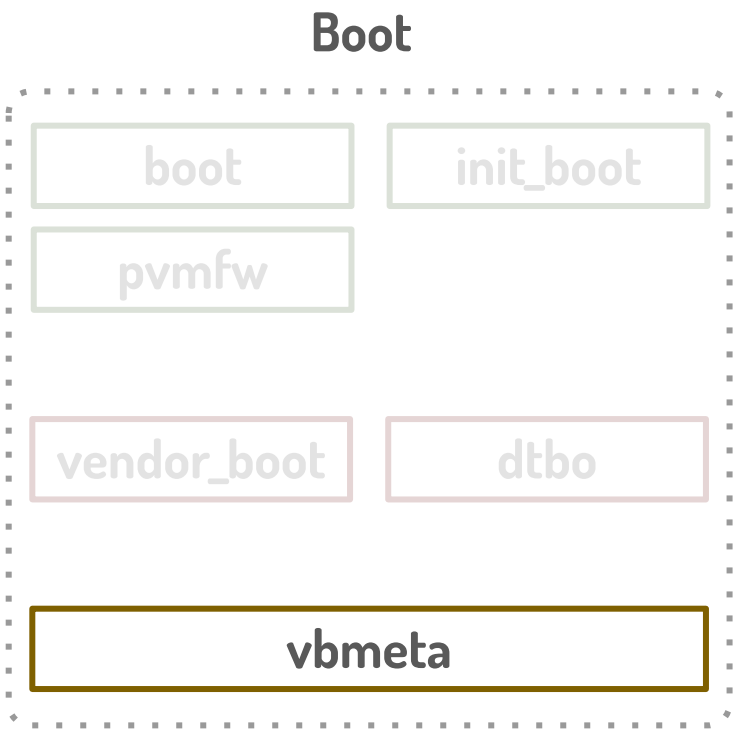
■ - Google provided. ■ - Mixed ownership. ■ - Vendor specific.

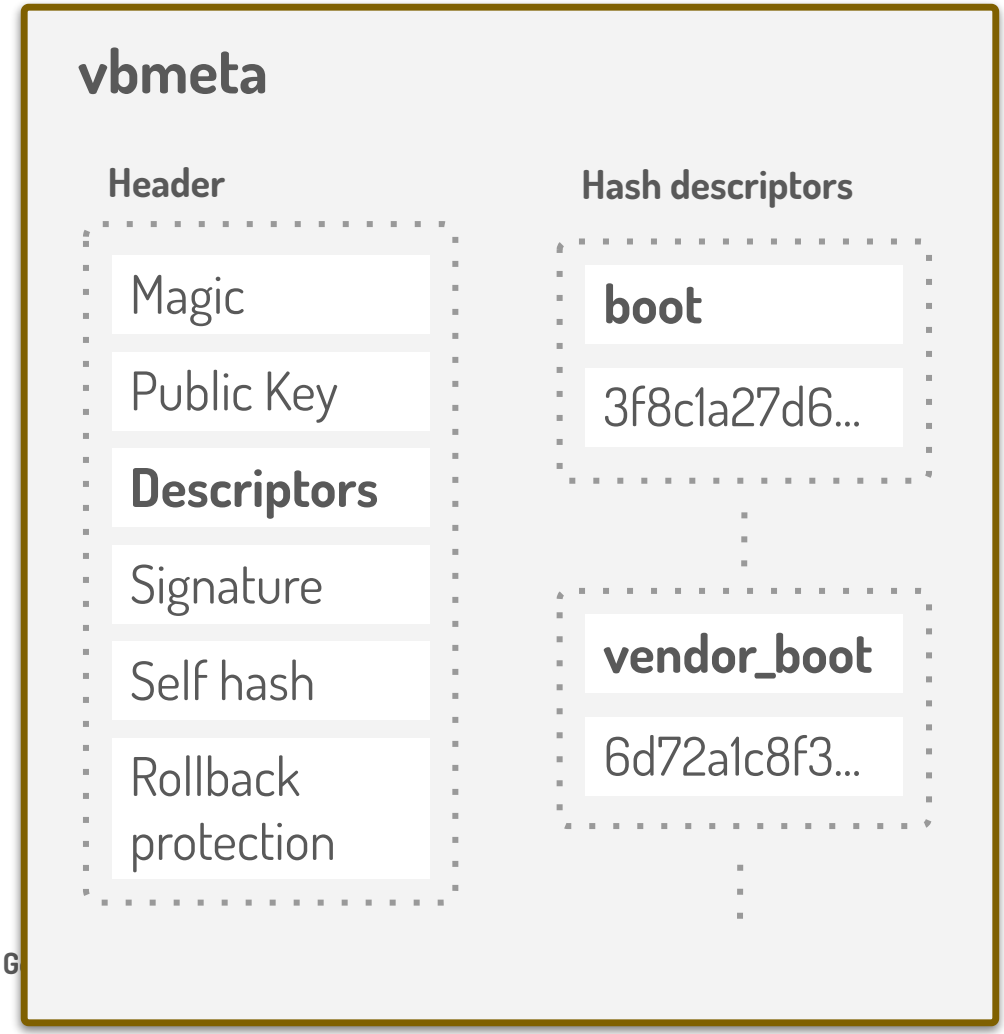
Android partitions



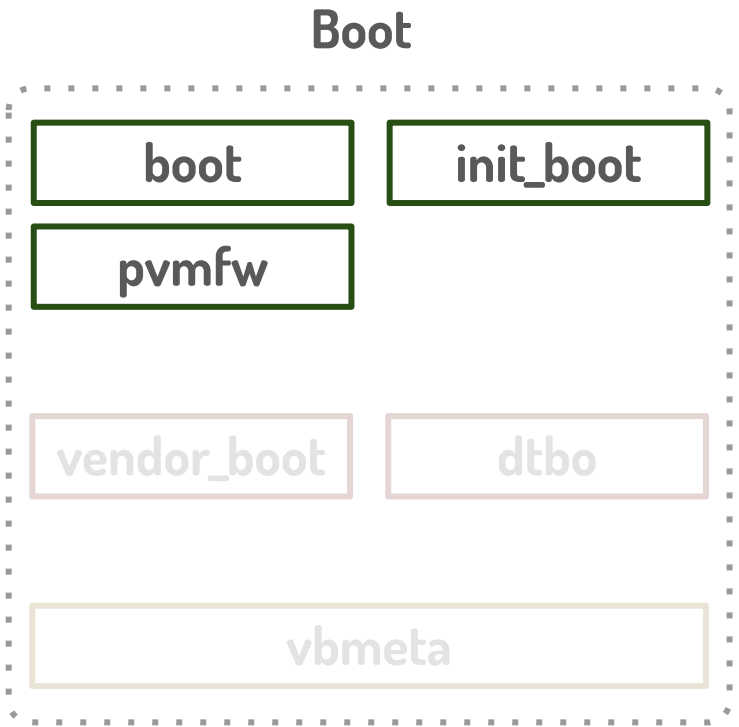
■ - Google provided. ■ - Mixed ownership. ■ - Vendor specific.

Android partitions





Android partitions



boot header

```
struct boot_img_hdr_v3 {
    uint8_t magic[BM_SIZE];
    uint32_t kernel_size;
    uint32_t ramdisk_size;
    uint32_t os_version;
    uint32_t header_size;
    uint32_t reserved[4];
    uint32_t header_version;
    uint8_t cmdline[N];
} __attribute__((packed));
```


■ - Google provided. ■ - Mixed ownership. ■ - Vendor specific.

Android partitions



vendor_boot header

```
struct vendor_boot_img_hdr_v4 {
    uint8_t magic[VBM_SIZE];
    uint32_t ramdisk_addr;
    uint32_t vendor_ramdisk_size;
    uint8_t cmdline[VB_ARGS_SIZE];
    uint32_t dtb_size;
    uint64_t dtb_addr;
    uint32_t vendor_ramdisk_table_size;
    uint32_t vendor_ramdisk_table_entry_num;
    uint32_t bootconfig_size;
    // ...
} __attribute__((packed));
```

- Google provided.
- Mixed ownership.
- Vendor specific.

Android partitions



dt_table header

```
struct dt_table_header {
  // ...
  uint32_t dt_entry_size;
  uint32_t dt_entry_count;
  uint32_t dt_entries_offset;
};
struct dt_table_entry_v1 {
  uint32_t dt_size;
  uint32_t dt_offset;
  uint32_t id;
  uint32_t rev;
  uint32_t flags;
  uint32_t custom[3];
};
```

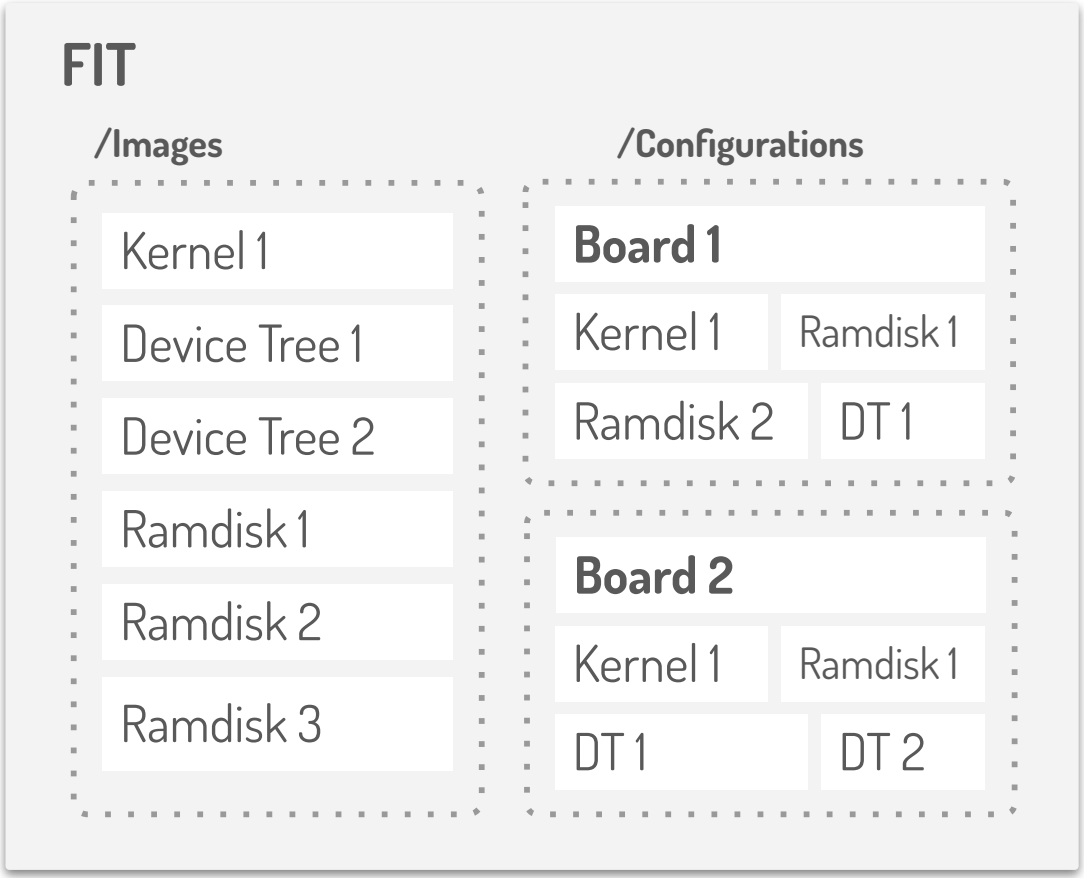
Problems

1. Using partition table to represent ownership model is too expensive.
    a. Some vendors introduced additional partitions to better represent their dev flow. E.g **vendor_kernel_boot** partition to split Kernel-related DTBs and ramdisks to be flashed separately.
2. Components (DTs, RAMDISKs) selection is limited.
    a. **dt_table** provides only few bytes of selection metadata which is not enough, so Android bootloader has to rely on DT internals (board-id, soc-id) to select at run-time.
    b. **dt_table** doesn't support images alignment, so DTs images aren't 8-bytes aligned on disk which costs additional relocations.
    c. AVB has to load/verify components even if they're not selected.

Introduce FIT

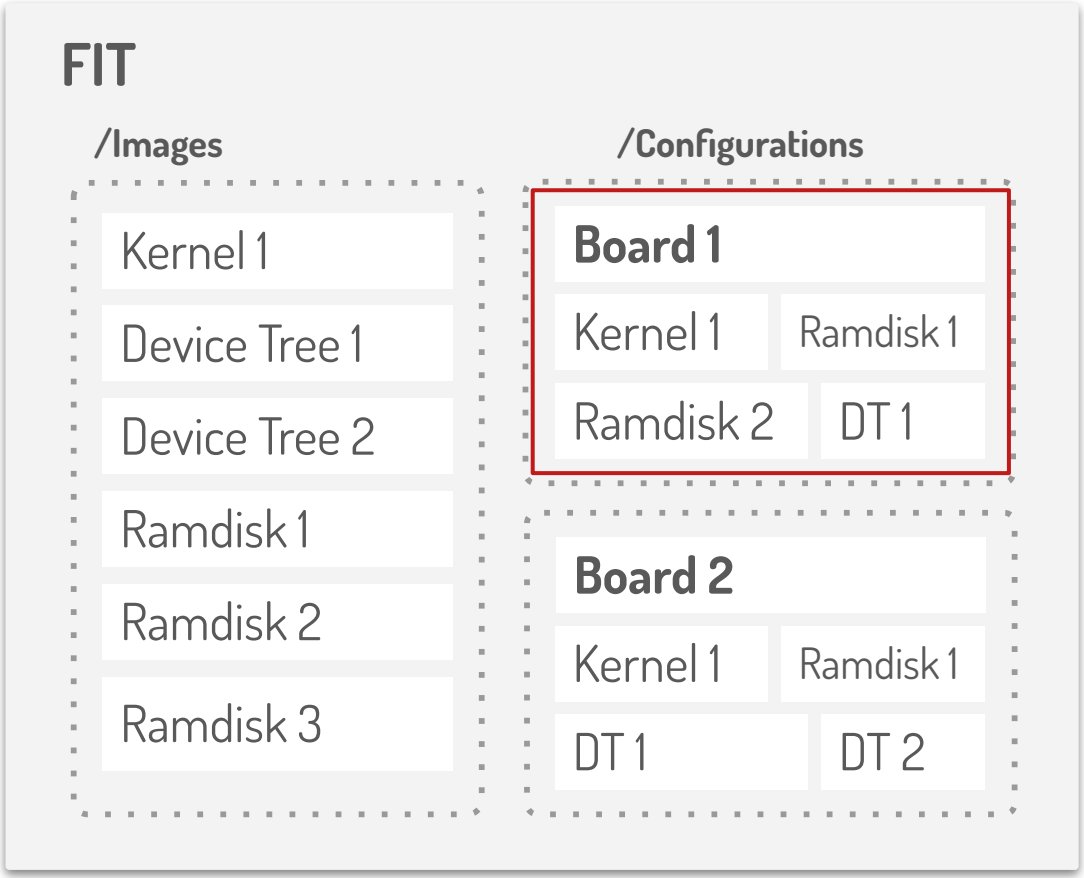
FIT (Flattened Image Tree) - universal FDT-based format to deliver boot artifacts.

- Proposed in 2008 by U-boot maintainers
- Published under Apache 2.0
- Used by Chrome OS



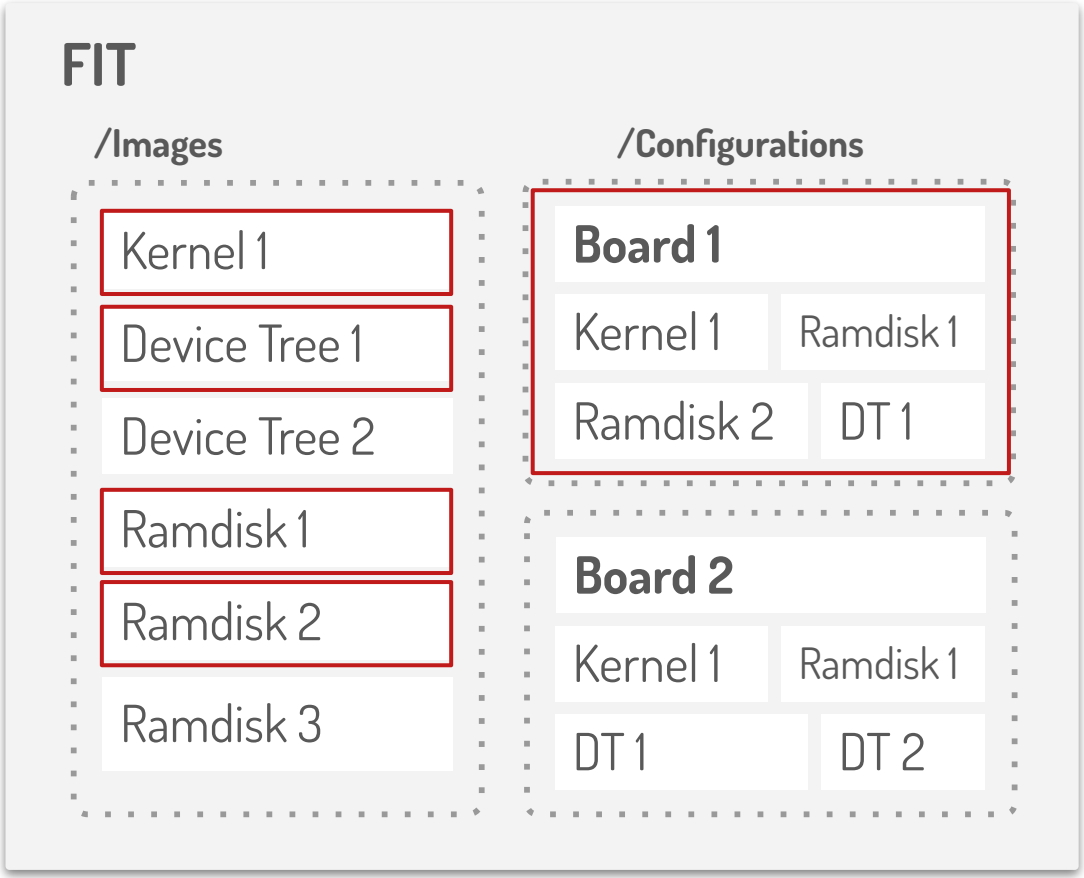
Introduce FIT

```
/dts-v1/;
/configurations/board-1 {
    compatible = "soc-1",
                 "board-1";
    kernel = "kernel-1";
    ramdisk = "ramdisk-1",
              "ramdisk-2";
    fdt = "fdt-1";
};
```



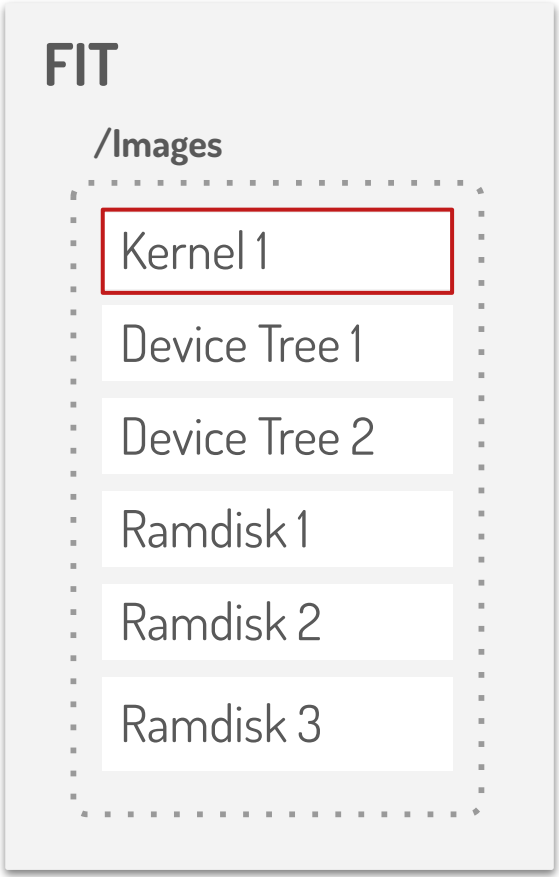
Introduce FIT

```
/dts-v1/;
/configurations/board-1 {
    compatible = "soc-1",
                 "board-1";
    kernel = "kernel-1";
    ramdisk = "ramdisk-1",
              "ramdisk-2";
    fdt = "fdt-1";
};
```



Introduce FIT

```
/dts-v1/;
/images/kernel-1 {
  description = "android16-6.18";
  data = /incbin/("./kernel");
  type = "kernel";
  os = "linux";
};
```



FIT features

1. External data. By default all images are embedded into FIT fdt. Can be overridden, so images will be appended at specific offset instead.
2. Compression (bzip2, gzip, lz4, etc).
3. Configurations/Images signature/hash/CRCs for integrity checks.

GBL's Initial FIT support

1. FIT image can be provided by **dtbo** partition.
    a. Only device trees and overlays are supported.
        i. If present, device trees from **vendor_boot** are **ignored**.
        ii. Kernel, ramdisk, command line, bootconfig are still obtained using Android boot structures.
    b. Device trees must be 8 bytes aligned external images to achieve better performance.
2. GBL's **OSConfigurationProtocol.SelectFitConfiguration** is used to delegate DT configuration selection to UEFI FW.

GBL's Initial FIT support

```
typedef struct GblEfiOsConfigurationProtocol {
  uint64_t revision;
  // ...
  EfiStatus (*select_fit_configuration)(
    struct GblEfiOsConfigurationProtocol* self,
    size_t fit_size, const uint8_t* fit,
    size_t metadata_size, const uint8_t* metadata,
    size_t* selected_configuration_offset);
};
```

Further ideas

1. Move vendor's ramdisks selection from vendor_boot into FIT
    a. Introduce VENDOR_FIT partitions to fully replace existing vendor_boot and dtbo
    b. Introduce FIT to replace all boot partitions?
2. FIT integration into Android tooling
    a. AOSP support
    b. AVB support, so multistage verification (FIT FDT, selected images)
    c. Flashing individual FIT images using Fastboot

Links

- Android partitions documentation
- GBL (Build artifacts, Development, Updates)
- GBL FIT documentation
- GBL OsConfigurationProtocol spec
- FIT spec

Reach us out: android-gbl@google.com

Thank you. Questions?